

- 11** A 0.483 g sample of glycine ($M_r = 75.0$) was placed in a bomb calorimeter and then ignited in the presence of excess oxygen. The temperature rose by $0.54\text{ }^{\circ}\text{C}$.

In a separate experiment using the same calorimeter, the combustion of 0.986 g of benzoic acid ($M_r = 122.0$) gave a temperature rise of $2.14\text{ }^{\circ}\text{C}$. The enthalpy change of combustion of benzoic acid is -3054 kJ mol^{-1} .

What is the enthalpy change of combustion, in kJ mol^{-1} , of glycine?

- A** -615 kJ mol^{-1}
- B** -967 kJ mol^{-1}
- C** -2423 kJ mol^{-1}
- D** -3812 kJ mol^{-1}